

PK SOMPURA – COMPLETE WEBSITE BLUEPRINT

This document contains the full architectural, design, technical, and strategic plan for building the PK Sompura temple construction website. It is written so a new development session can begin without re-explaining prior discussions.

1. PROJECT IDENTITY

Business Type: Temple Construction (Joint Family Business) Name: PK Sompura Structure: Grandfather + Father + Uncles (Collective Legacy) Next Generation: Not publicly featured

Purpose Hierarchy: 1. Trust & credibility for serious clients 2. Legacy preservation 3. Public awareness 4. SEO visibility 5. Professional presentation in pitches

Tone Direction: - Architectural & cultural (NOT devotional-heavy) - Dignified, restrained, timeless - Silent authority

Temples to Feature: Maximum 10 (curated, not a list of 50+)

Languages: English + Gujarati + Hindi

2. EXPERIENCE PHILOSOPHY

Website Type: Single-page cinematic scroll experience. No grids. No lists. No tables. No gallery layouts.

Scroll Model: - Primarily vertical scroll - Pinned cinematic sections - Selective horizontal reveals for special narrative moments

Core Design Metaphor: Website behaves like entering and walking through a temple.

Golden Rule: Show mastery without teaching the method.

3. ENTRY EXPERIENCE (CUSTOM GATE)

First Visit: - Stone-textured dark background - Minimal text: PK Sompura – Sacred Architecture - “Enter” button - Abstract ancient stone doors - Slow, restrained opening animation - Light revealed behind doors

Repeat Visits: - Gate auto-skipped - Optional “Enter Ritual” button available

Tracking: - Anonymous localStorage flag for first-time detection - No login - No email collection

Optional: - Subtle ambient sound (OFF by default)

4. FULL PAGE STRUCTURE

SECTION 1 – Threshold - Minimal branding - Slow fade-in - Establish silence and tone

SECTION 2 – First Reveal (Pinned) - Background temple texture - Foreground architectural cutouts - Minimal layered text

SECTION 3 – Architectural Milestone (Vaishnodevi Mandir – Ahmedabad) - Pinned cinematic reveal - Partial angle (no elevation view) - Text: first of its kind in India - Subtle horizontal reveal transitioning to: - Bhavnagar (Gujarat) - Gulbarga (Karnataka)

SECTION 4 – Featured Temple Journey - 7–10 temples appear as chapters - Each temple includes: - 1 hero image (angled, safe framing) - 2–3 supporting layered visuals (optional) - City + Year - One defining architectural statement - No floor plans - No elevation symmetry

SECTION 5 – Craft & Knowledge - Stone textures - Macro fragments - Text about proportion, tradition, adaptation - No technical instruction

SECTION 6 – Scale Without Exposure - Statement: “More than 50 sacred structures across regions.” - Fast partial flashes (blurred / cropped) - No listing

SECTION 7 – Contact & Inquiry - Professional contact form - No aggressive CTA - Business-focused tone

5. VISUAL SYSTEM

Mood: Stone + Shadow + Silence With Light as Revelation

Color Direction: - Deep charcoal - Stone gray - Muted sandstone - Soft warm light accents

Typography Direction: - Serif for headings (timeless, architectural) - Clean sans-serif for body - Multilingual compatible fonts

Motion Rules: - Slow easing - No aggressive snapping - No flashy effects - Parallax subtle

6. IMAGE & CUTOUT STRATEGY

Layered System: 1. Background layer – full image (darkened / blurred) 2. Mid layer – architectural cutouts (pillars, arches, shikhara fragments) 3. Foreground layer – text elements

Image Security: - No orthographic elevation views - No floor plans - No structural construction exposure - Avoid perfectly centered symmetry - Controlled resolution

Optional Enhancements: - Texture overlays - Mask transitions - Soft depth parallax

7. TECH STACK (PYTHON-CENTERED)

Backend: - FastAPI - PostgreSQL database

Frontend: - React or Next.js - GSAP + ScrollTrigger for animations - Tailwind CSS for styling

Multilingual Handling: - Store translations in database - API language parameter - Structured language routing

Analytics: - Plausible or Vercel Analytics - Anonymous visit tracking - Scroll depth monitoring

Deployment: Option A: - Frontend → Vercel - Backend → Railway / Render

Option B: - Full-stack deployment on VPS (DigitalOcean / AWS)

8. DATA STRUCTURE (SIMPLIFIED)

Temple Model: - id - name (en, gu, hi) - city - state - year - short_description (3 languages) - milestone_flag - order_index

Image Model: - id - temple_id - image_url - cutout_flag - display_order

9. SEO STRATEGY

Target Keywords: - Temple construction India - Temple contractor Gujarat - Mandir architecture Ahmedabad

Implementation: - Metadata per language - Schema markup - Fast page load - Proper heading hierarchy

10. SECURITY STRATEGY

Protection Method: - Composition-based protection - Layered visuals - Motion-based reveal - No technical documentation public

No reliance on: - Right-click blocking - Watermarks

11. OPTIONAL FEATURES (FUTURE EXPANSION)

- Professional photography upgrade
 - Private client portal
 - Press/media section
 - Awards recognition
 - Limited access blueprint sharing (private)
 - Documentary-style short video integration
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12. DEVELOPMENT PHASE PLAN

Phase 1 – Foundation - Backend API setup - Database schema - Basic frontend layout

Phase 2 – Core Scroll Experience - Gate animation - Pinned sections - First temple milestone

Phase 3 – Temple Chapters - Add remaining featured temples - Integrate cutout layering

Phase 4 – Multilingual Support - Translation integration - Language switching

Phase 5 – SEO + Performance Optimization - Metadata - Lazy loading - Compression

Phase 6 – Deployment & Testing - Cross-device testing - Performance audits - Final polish

IMPLEMENTATION-GRADE TECHNICAL SPECIFICATION

This section upgrades the blueprint into execution-level detail.

13. UI SYSTEM SPECIFICATION

13.1 Layout Grid

Base Grid: - 12-column responsive grid - Max width: 1440px - Content width: 1200px - Side padding: 5–8vw (responsive)

Vertical Rhythm: - Base spacing unit: 8px - Section padding: 96px desktop / 64px tablet / 48px mobile

Pinned Section Height: - 200vh–300vh per cinematic section - Content stays pinned while scroll animates layers

14. COLOR SYSTEM

Primary Background: - Deep Charcoal: #0F0F10

Stone Base: - Dark Stone: #1C1C1E - Muted Slate: #2A2A2D

Accent Light (Revelation Tone): - Warm Sand Glow: #D6B98C - Soft Gold Hint: #C5A45F (used sparingly)

Text Colors: - Primary: #E8E6E3 - Secondary: #B5B3AE

Rule: Accent color must never exceed 5% of screen usage.

15. TYPOGRAPHY SYSTEM

Heading Font (Serif, timeless): - Playfair Display OR Cormorant Garamond

Body Font (Clean Sans): - Inter OR Source Sans 3

Gujarati & Hindi Support: - Noto Serif Gujarati - Noto Serif Devanagari

Hierarchy: H1 – 56px H2 – 40px H3 – 28px Body – 18px Small Meta – 14px

Letter spacing for headings: +0.5px Line height body: 1.6

16. GATE ANIMATION SPECIFICATION

16.1 Structure

HTML Layers: - GateContainer - LeftDoor - RightDoor - LightLayer - EnterButton

16.2 Animation Timing (GSAP)

Phase 1 – Fade In (1.5s) - Opacity 0 → 1

Phase 2 – Button Press Trigger - 300ms delay

Phase 3 – Door Opening (2.8s) - LeftDoor: translateX(-110%) - RightDoor: translateX(110%) - Ease: power3.inOut

Phase 4 – Light Expansion (1.8s overlapping) - LightLayer scale 1 → 1.3 - Opacity increase 0.3 → 0.8

Phase 5 – Transition to Hero - Gate container fade out 1s - Scroll locked during animation

Repeat Visit Logic: - Check localStorage.hasVisited - If true → skip to Hero

17. SCROLL ENGINE STRUCTURE

Using GSAP ScrollTrigger

Main Timeline: - Each major section has its own timeline - Use pin: true for milestone sections

Example: ScrollTrigger.create({ trigger: sectionRef, start: "top top", end: "+=200%", pin: true, scrub: 1 })

Parallax Ratios: - Background: 0.3 speed - Cutouts: 0.6 speed - Foreground text: 1.0

18. CUTOUT PRODUCTION WORKFLOW

Step 1 – Select safe angled image Step 2 – Remove background (Photoshop / remove.bg) Step 3 – Manually refine edges Step 4 – Export WebP with transparency Step 5 – Optimize resolution (max 1800px width) Step 6 – Upload to Cloudinary

Never use: - Perfect front elevation - Full symmetric capture - Blueprint-like imagery

19. BACKEND IMPLEMENTATION PLAN (FASTAPI)

19.1 Project Structure

backend/ main.py models.py schemas.py database.py routes/ temples.py contact.py

19.2 Core Endpoints

GET /temples?lang=en GET /temples/featured POST /contact GET /analytics-summary (admin only)

19.3 Database Schema

Temple Table: - id - name_en - name_gu - name_hi - city - state - year - description_en - description_gu - description_hi - is_milestone - order_index

Image Table: - id - temple_id - url - is_cutout - order_index

20. FRONTEND FOLDER STRUCTURE

frontend/ components/ Gate.jsx Hero.jsx TempleMilestone.jsx TempleChapter.jsx CraftSection.jsx ContactSection.jsx animations/ gateAnimation.js scrollAnimations.js hooks/ utils/

21. PERFORMANCE OPTIMIZATION

- Use lazy loading for images
- Use dynamic import for heavy animation sections
- Compress all images to WebP
- Defer non-critical scripts
- Use prefers-reduced-motion media query

Target Lighthouse Score: 90+

22. DEVELOPMENT PHASE BREAKDOWN (SELF-BUILD)

Week 1: - Setup FastAPI - Setup PostgreSQL - Basic API endpoints

Week 2: - Setup React/Next frontend - Basic layout without animations

Week 3: - Implement Gate animation - Add first pinned section

Week 4: - Add Temple milestone section - Integrate cutouts

Week 5: - Add multilingual system - Connect API fully

Week 6: - Performance optimization - SEO metadata - Deployment

END OF MASTER BLUEPRINT – IMPLEMENTATION VERSION

23. CURRENT IMPLEMENTATION STATUS (WHAT HAS BEEN COMPLETED)

This section documents the exact current progress so development can continue seamlessly in any environment or with any assistant.

Repository

GitHub repository created and connected:

Repository Name: PK-Sompura

Local Path: E:\Projects\PK-Sompura

Repository Structure:

PK-Sompura/ ├── backend/ | ├── main.py | ├── database.py | ├── models.py | ├──
schemas.py | ├── routes/ ├── frontend/ ├── venv/ ├── requirements.txt ├── .gitignore

Git initialized, connected, and pushed successfully.

Backend Status

Environment: Python virtual environment created and activated.

Installed packages: - fastapi - uvicorn - sqlalchemy - psycopg2-binary - python-dotenv

Backend server working and verified.

Current backend code:

backend/main.py

```
from fastapi import FastAPI

app = FastAPI()

@app.get("/")
def root():
    return {"message": "PK Sompura Backend Running"}
```

Backend runs successfully using:

uvicorn backend.main:app --reload

Verified endpoint: http://127.0.0.1:8000

Frontend Status

Frontend initialized using:

React + Vite

Installed packages: - gsap - axios

Development server working and verified.

Runs using:

npm run dev

Verified at: <http://localhost:5173>

Gate System Status (Fully Implemented)

Gate component created:

frontend/src/components/Gate.jsx frontend/src/components/Gate.css

Functional features implemented:

- Gate appears on initial load
- Two-door system implemented
- GSAP animation timeline created
- Doors rotate in 3D
- Doors now open inward (correct temple entry behavior)
- Light layer behind gate implemented
- Gate fades out after opening

LocalStorage repeat-visit logic temporarily disabled for development.

Current animation sequence:

1. User clicks Enter button
2. Camera subtle push-in effect
3. Left door rotates inward using rotationY positive
4. Right door rotates inward using rotationY negative
5. Light layer opacity increases
6. Gate container fades out
7. Main site content becomes visible

Perspective system implemented using:

CSS perspective property

GSAP used for animation timeline control.

Frontend Component Structure

frontend/src/

|— components/ | |— Gate.jsx | |— Gate.css |— App.jsx |— main.jsx

Gate integrated into App.jsx.

Development Mode Adjustments

LocalStorage gate-skip logic disabled to allow repeated animation testing.

Will be re-enabled in production.

24. NEXT DEVELOPMENT TASKS (IN ORDER)

These tasks are required to complete the full cinematic site.

Phase 1 – Gate Visual Upgrade

Required tasks:

- Add realistic stone texture overlay
- Add hinge depth illusion
- Add door thickness illusion
- Add interior corridor or glow environment

Optional tasks:

- Add ambient sound
 - Add camera depth movement
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Phase 2 – Core Scroll Architecture

Implement scroll sections using GSAP ScrollTrigger.

Sections to create:

- Hero Section
- Vaishnodevi Milestone Section
- Temple Chapter Sections (max 10)
- Craft Section
- Contact Section

Each section will use:

Pinned scroll sections Layered cutout images Parallax depth layers

Phase 3 – Cutout Image System

For each temple:

Create:

- Background image
- Cutout foreground element
- Light and shadow layers

Cutouts created manually using image editing tools.

Images optimized before upload.

Phase 4 – Backend Data Integration

Implement PostgreSQL database.

Create tables:

Temple TempleImage

Create FastAPI endpoints:

GET /temples GET /temples/featured POST /contact

Frontend fetches temple data dynamically.

Phase 5 – Multilingual Support

Add language toggle.

Backend stores:

English Gujarati Hindi

Frontend loads language-specific content.

Phase 6 – Deployment

Backend deployment options:

Railway Render VPS

Frontend deployment:

Vercel

Domain connection.

25. DESIGN PRINCIPLES (STRICT RULES)

Must follow:

- No list-based temple display
- No grid gallery
- No blueprint-style images
- No elevation symmetry exposure

Must use:

- Cinematic scroll experience
- Layered cutout depth system
- Controlled architectural reveal

Goal:

Digital architectural legacy presentation.

26. FINAL OBJECTIVE

The final website must:

- Present PK Sompura temple construction legacy
 - Provide immersive cinematic experience
 - Protect architectural intellectual property
 - Provide client credibility
 - Be scalable and maintainable
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END OF MASTER BLUEPRINT – FULL IMPLEMENTATION AND CURRENT STATE DOCUMENT

This document defines the complete vision, structure, and technical plan required to build the PK Sompura website independently without needing prior conversation context.